

Freight Rate Prediction

Machine Learning Assessment Report

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1. Approach Summary

The objective was to predict freight rates for unseen loads using 48,000 labeled historical shipments. The solution followed a time-aware development process: exploratory analysis and data-quality checks, chronological validation, comparison of several regression approaches, error analysis, and final retraining on the complete labeled January–October 2025 dataset.

XGBoost was selected as the final model. On the September–October temporal holdout it achieved an MAE of \$109.12, MAPE of 4.79%, RMSE of \$634.02, and R^2 of 0.827. The model was then retrained on all available labeled data for the final November–December predictions.

2. Data Exploration and Quality

The development dataset contains 48,000 rows and covers January 1 through October 31, 2025. The target, `posted_rate`, is right-skewed: its mean is \$2,373.98, median is \$2,030.76, and maximum is \$25,533. Distance is the strongest individual numerical predictor, with a Pearson correlation of 0.91 with `posted_rate`.

Data-quality issue	Observed	Treatment
Missing weight	300 rows (0.625%)	Median imputation fitted on training data
Missing market_index	374 rows (0.779%)	Median imputation fitted on training data
Negative weight	292 rows	Absolute value; magnitude retained

Negative weights ranged down to -47,500 lb while positive weights reached +47,500 lb, supporting the interpretation that these values are sign-entry errors rather than a separate physical scale. No rows were removed for these issues.

3. Validation Strategy

A chronological split was used instead of a random split because the final task is explicitly future prediction. The labeled data ends in October, while the final validation data contains 12,000 unique loads from November 1 through December 31, 2025. Randomly mixing earlier and later observations could therefore produce an overly optimistic estimate of future performance.

Subset	Period	Rows	Share
Training	Jan 1 – Aug 31, 2025	38,477	80.2%
Validation	Sep 1 – Oct 31, 2025	9,523	19.8%

Preprocessing statistics were fitted only on the January–August training portion and then reused unchanged on the September–October holdout, preventing leakage. Equipment composition remained stable across the split, while `market_index` showed temporal movement, further supporting a time-based validation design.

4. Features and Model Development

The final common feature set uses pickup, delivery, distance, equipment, weight, and date. Calendar features (month, day of week, day of month, and days since January 1, 2025) are derived from date. Pickup, delivery, and equipment are one-hot encoded for XGBoost.

A common feature set was preferred because the required fixed December input does not contain coordinates, market_index, or quote_signal. An extended CatBoost experiment using those additional fields did not improve temporal validation performance. Additional route, distance-band, and interaction features were also tested with XGBoost, but increased MAE from \$109.12 to \$112.88, so the simpler feature set was retained.

5. Model Comparison

Model	MAE	RMSE	R ²
Median baseline	1,148.924	1,569.424	-0.058
Distance linear regression	196.950	654.416	0.816
Extra Trees	165.504	691.846	0.794
CatBoost (common)	112.824	635.093	0.827
CatBoost (log target)	112.151	634.938	0.827
CatBoost (extended)	116.455	636.398	0.826
XGBoost	109.118	634.018	0.827

The distance-only linear model already performed strongly, confirming that freight pricing is largely mileage-driven. Gradient-boosting models provided the best results by learning smaller adjustments around this relationship. XGBoost achieved the lowest MAE and was therefore selected.

6. Error Analysis

XGBoost improved prediction accuracy across the main body of the validation distribution: median absolute error was \$35.77, the 90th percentile was \$134.89, and the 95th percentile was \$190.16. However, the maximum error remained above \$16,000. The same extreme-rate observations were also among the worst CatBoost errors.

This indicates that the high RMSE is driven by a small group of unusually expensive loads rather than broad systematic error. Their observed rates are far above the values suggested by the available shipment characteristics. Because the pattern persisted across model families and target transformation, these cases were retained rather than removed as outliers.

7. Final Model and Training

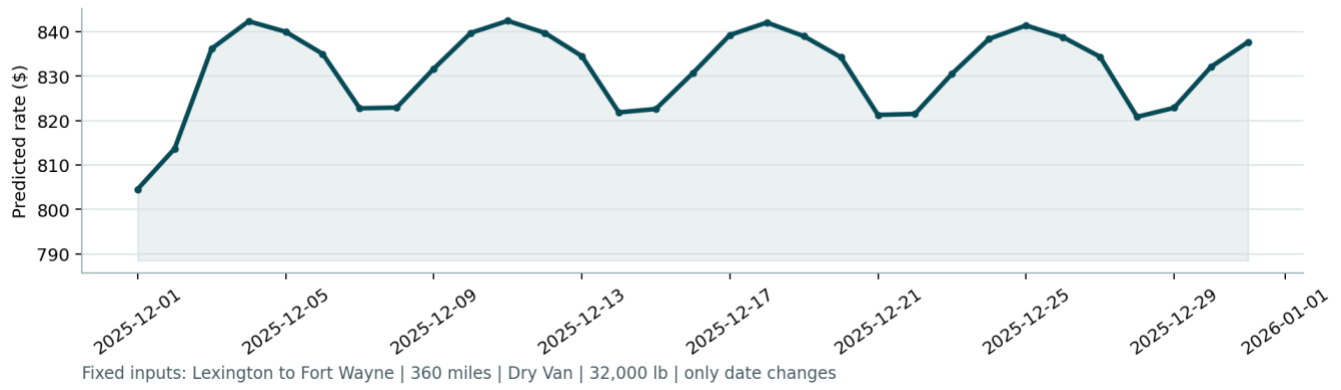
The final model is an XGBoost regressor using the original common feature set. After model selection, the pipeline was retrained on the complete January–October development dataset so that all labeled information was available for the final predictions. The validation holdout was used only for development decisions and was not held back from final training.

Final temporal holdout performance: MAE \$109.12 | MAPE 4.79% | RMSE \$634.02 | R² 0.827

8. Fixed December Prediction Chart

The assessment requires the fixed December chart generated by the provided score.py after final predictions are created.

Candidate: December 2025 Predicted Load Rate



9. Conclusion

The final solution uses a leakage-safe temporal validation strategy and a compact, reproducible XGBoost pipeline. The model substantially outperforms simple baselines and achieves a 4.79% MAPE on the future-period holdout. Further feature expansion was not pursued because tested additions did not improve validation performance and the remaining large errors were concentrated in a small set of extreme-rate observations.